HUMAN ALBUMIN

Blood product — consent & traceability

20% for paracentesis

CHEMICAL NAME Human serum albumin (single polypeptide of 585 amino acids, ~66.5 kDa) from pooled donor plasma — 4–5% (isotonic) or 20–25% (hyperoncotic)

CLASS Colloid — plasma-derived blood product

TRADE NAMES Albumex · Human Albumin Grifols · Alburex · Zenalb

THERAPEUTIC USES / INDICATIONS

- Large-volume paracentesis (>5 L) in cirrhosis — prevents post-paracentesis circulatory dysfunction.
- Spontaneous bacterial peritonitis and hepatorenal syndrome (with terlipressin).
- Hypovolaemia when crystalloids insufficient (4–5%); plasma exchange.
- Burns (after 24 h, selected); hypoalbuminaemia with diuretic-resistant oedema (limited evidence).

ADMINISTRATION

- **IV** via standard giving set; no filter needed; do not dilute 20% with water (haemolysis).
- Use within 3–4 h of opening; check expiry, clarity, batch number.
- Record batch number (traceability); consent as a blood product where required.

DOSAGES

- **Paracentesis:** 8 g albumin per litre drained (e.g. 100 mL of 20% per 2.5 L removed).
- **SBP:** 1.5 g/kg on day 1, 1 g/kg on day 3.
- **AKI/HRS in cirrhosis:** 1 g/kg/day for 2 days, then (with terlipressin) 20–40 g/day.
- **Hypovolaemia (adult):** initial 25 g, repeated if response inadequate after 15–30 min.

PHARMACOKINETICS

- Distributes in intravascular space initially; ~10% leaks within 2 h (more in sepsis/capillary leak).
- Catabolised slowly. **t½:** ~19 days.

PHARMACODYNAMICS

- Provides colloid oncotic pressure → holds/draws fluid into circulation; binds and transports drugs, bilirubin, hormones.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Flushing, fever, chills.
- Nausea.

ADVERSE EFFECTS (SERIOUS)

- Fluid overload, pulmonary oedema (20% solution).
- Hypersensitivity/anaphylaxis (rare).
- Infection transmission (theoretical; heat treated).

CONTRAINDICATIONS & CAUTIONS

- Severe heart failure; severe anaemia (risk of overload).
- Hypersensitivity to albumin.

KEY INTERACTIONS

- No significant drug interactions.

NURSING MANAGEMENT

- Check product, expiry, batch number; record in notes.
- Monitor BP, HR, RR, SpO_2, signs of overload.
- Paracentesis: give during/after drain as per protocol; monitor drain output and BP.
- Monitor albumin, renal function.

PATIENT EDUCATION

- Explain it is a blood-derived product and why it is needed.
- Report breathlessness, chest tightness, rash or chills.

KEY POINTS

- Colloid from human plasma (4–5% or 20%).
- SBP and hepatorenal syndrome.
- Paracentesis: 8 g per litre drained.
- Overload risk; traceability.

SOURCES CHECKED **[961]** EASL decompensated cirrhosis guideline (2018) **[381]** Evans et al., Intensive Care Med 2021 **[962]** Albuminex, IV (US label) **[963]** Albuminex, IV (US label, BPL)

Full references p. 432–457